

Program Evaluation PS1

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```
## load data
# @Jessica this data path will work if you set your directory to our program eval repo
df_data <- read_csv("ps1_data_spring26.csv")

## Rows: 500 Columns: 13
## -- Column specification -----
## Delimiter: ","
## dbl (13): farms_trt, farms_got_loan, baseline_livestock, baseline_hhsize, ba...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
#df_data <- read_csv("C:/Users/blueb/Downloads/ps1_data_spring26.csv", show_col_types = FALSE)

glimpse(df_data)

## Rows: 500
## Columns: 13
## $ farms_trt          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ farms_got_loan     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
## $ baseline_livestock <dbl> 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, ~
## $ baseline_hhsize    <dbl> 6, 2, 5, 3, 5, 5, 6, 6, 4, 6, 4, 6, 4, 5, 5, 4, 5, ~
## $ baseline_agexp     <dbl> 82, 84, 62, 88, 109, 34, 49, 71, 63, 82, 73, 80, 96~
## $ baseline_credit    <dbl> 51, 47, 55, 60, 91, 66, 40, 65, 60, 20, 34, 73, 63,~
## $ baseline_income    <dbl> 513, 472, 551, 601, 911, 661, 403, 656, 603, 201, 3~
## $ livestock_temp     <dbl> -0.3813980, 0.5124075, -0.1106800, -2.0293130, 0.93~
## $ endline_livestock  <dbl> 1, 1, 0, 0, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 1, 0, ~
## $ endline_hhsize     <dbl> 6, 2, 5, 3, 5, 5, 6, 6, 4, 6, 4, 6, 4, 5, 5, 4, 5, ~
## $ endline_agexp      <dbl> 67, 84, 44, 83, 105, 50, 51, 78, 68, 81, 66, 83, 10~
## $ endline_credit     <dbl> 51, 47, 59, 57, 93, 68, 37, 65, 60, 21, 34, 74, 62,~
## $ endline_income     <dbl> 686, 875, 2374, 1622, 2234, 876, 730, 2651, 2074, 2~

summary(df_data)

##   farms_trt   farms_got_loan baseline_livestock baseline_hhsize
##   Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :2.000
##   1st Qu.:0.000   1st Qu.:0.000   1st Qu.:0.000   1st Qu.:4.000
##   Median :1.000   Median :0.000   Median :1.000   Median :5.000
##   Mean   :0.502   Mean   :0.368   Mean   :0.582   Mean   :4.968
##   3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:6.000
##   Max.   :1.000   Max.   :1.000   Max.   :1.000   Max.   :8.000
##   baseline_agexp baseline_credit baseline_income livestock_temp
##   Min.    : 2.00   Min.    : 8.00   Min.    : 86.0   Min.    :-2.75567
##   1st Qu.: 60.00   1st Qu.: 45.00   1st Qu.: 450.0   1st Qu.: -0.72171
```

```
## Median : 75.00    Median : 54.00    Median : 543.0    Median :-0.08578
## Mean   : 73.93    Mean   : 54.48    Mean   : 547.2    Mean   :-0.03103
## 3rd Qu.: 87.00    3rd Qu.: 64.00    3rd Qu.: 643.0    3rd Qu.: 0.65427
## Max.   :138.00    Max.   :100.00    Max.   :1003.0    Max.   : 2.49916
##  endlivestock endlivestock endlivestock endlivestock
## Min.   :0.00      Min.   :2.000    Min.   : 0.0      Min.   : 0.00
## 1st Qu.:0.00      1st Qu.:4.000    1st Qu.: 69.0     1st Qu.: 36.00
## Median :1.00      Median :5.000    Median : 91.0     Median : 47.00
## Mean   :0.67      Mean   :4.968    Mean   : 626.2     Mean   : 47.22
## 3rd Qu.:1.00      3rd Qu.:6.000    3rd Qu.:1563.0    3rd Qu.: 59.00
## Max.   :1.00      Max.   :8.000    Max.   :1653.0    Max.   :103.00
##  endlivestock endlivestock endlivestock endlivestock
## Min.   : 228
## 1st Qu.:1197
## Median :2272
## Mean   :2896
## 3rd Qu.:4730
## Max.   :7346
```

1. The impact of access to microcredit for a household i on income is the difference between the household's income with and without access to microcredit.

In math: $T_i = Y_i(1) - Y_i(0)$, where T_i is the impact of treatment for household i in \$/month, $Y_i(1)$ is the income of household i with access to microcredit, and $Y_i(0)$ is the income of household i without access to microcredit.

2. Measuring household specific treatment effects is impossible due to the fundamental problem of causal inference as we are unable to observe both potential outcomes of a household being treated and untreated at the same time. We are unable to observe the counterfactual, and are therefore unable to calculate individual household effects.
3. This estimator measures not just the average treatment effect on income, but the income impact of other variables that distinguish households with microcredit access from households without microcredit access. Their suggested estimator is sometimes called the “naive estimator” because of the selection problem; access to microcredit is likely not provided at random, which means that there are other factors that may bias the estimate, unequally impacting income for those with access to microcredit and those without. For example, households with access to microfinancing might tend to have more resourced local governments, which assist household income with a variety of interventions in addition to microcredit options. This disparity, among others, would bias the estimate.
4. $ATT = E[Y_i(1) - Y_i(0)|D_i=1]$ In this context, the ATT would be the average outcome from receiving the microcredit on households who did receive the microcredit subtracted by the average outcome from not receiving the microcredit on the households who did receive the microcredit. The FARMS data will not allow us to estimate the ATT, because we are unable to observe the outcomes of households who received the microcredit had they not received the microcredit.
5. The histograms suggest that treated households have a higher average income pre-treatment. The balance table shows that the treated and untreated groups were statistically different on income, agricultural expenditure, household size, and livestock ownership even before treatment (access to microcredit for the treatment group). In order to estimate the causal effect of `farms_trt`, we must be able to reasonably assume that the treatment and control group are statistically the same on unobserved characteristics.

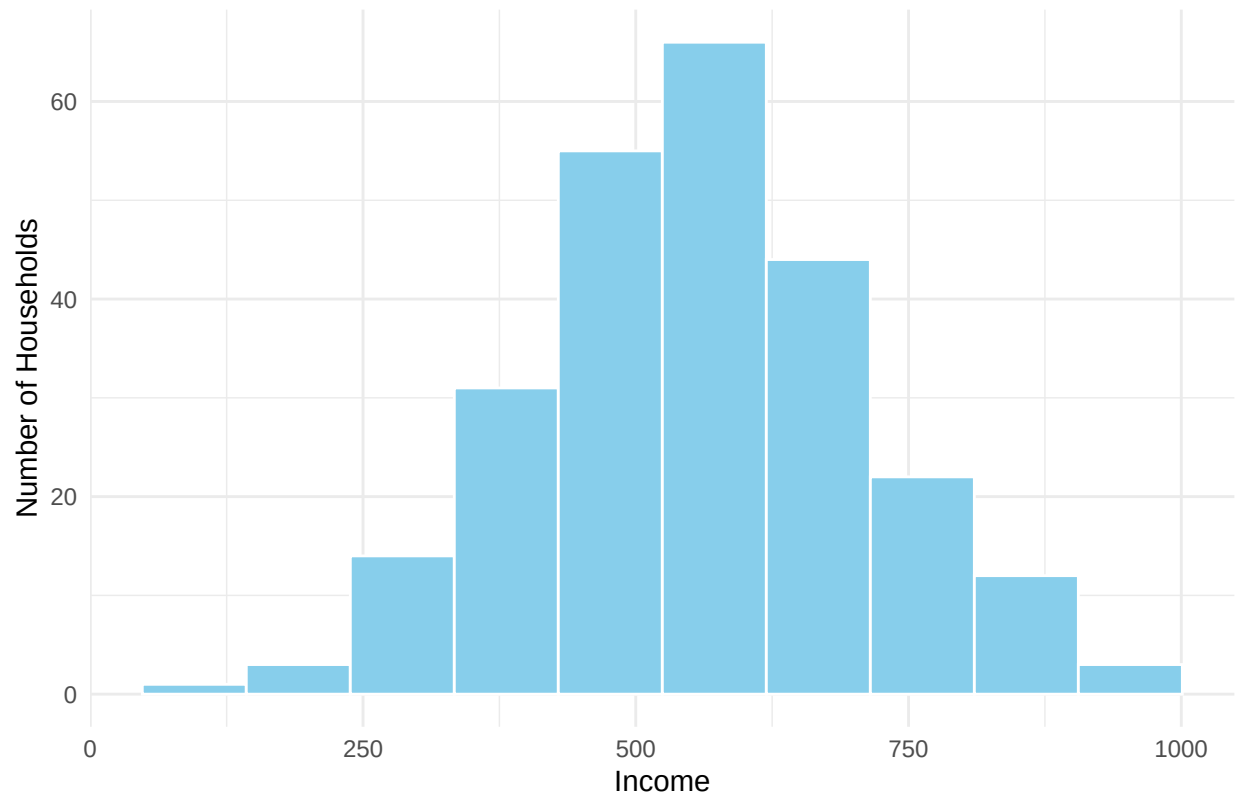
```
# HISTOGRAMS
# create histogram of pre-treatment incomes

# define an object that holds baseline income for treated households
```

```
treated_inc_pre <- df_data$baseline_income[df_data$farms_trt == 1]

# plot income of treated households
ggplot(data.frame(treated_inc_pre), aes(x = treated_inc_pre)) +
  geom_histogram(fill = "skyblue", color = "white", bins = 10) +
  labs(title = "Pre-Treatment Income on Treated Households",
       x = "Income",
       y = "Number of Households") +
  theme_minimal()
```

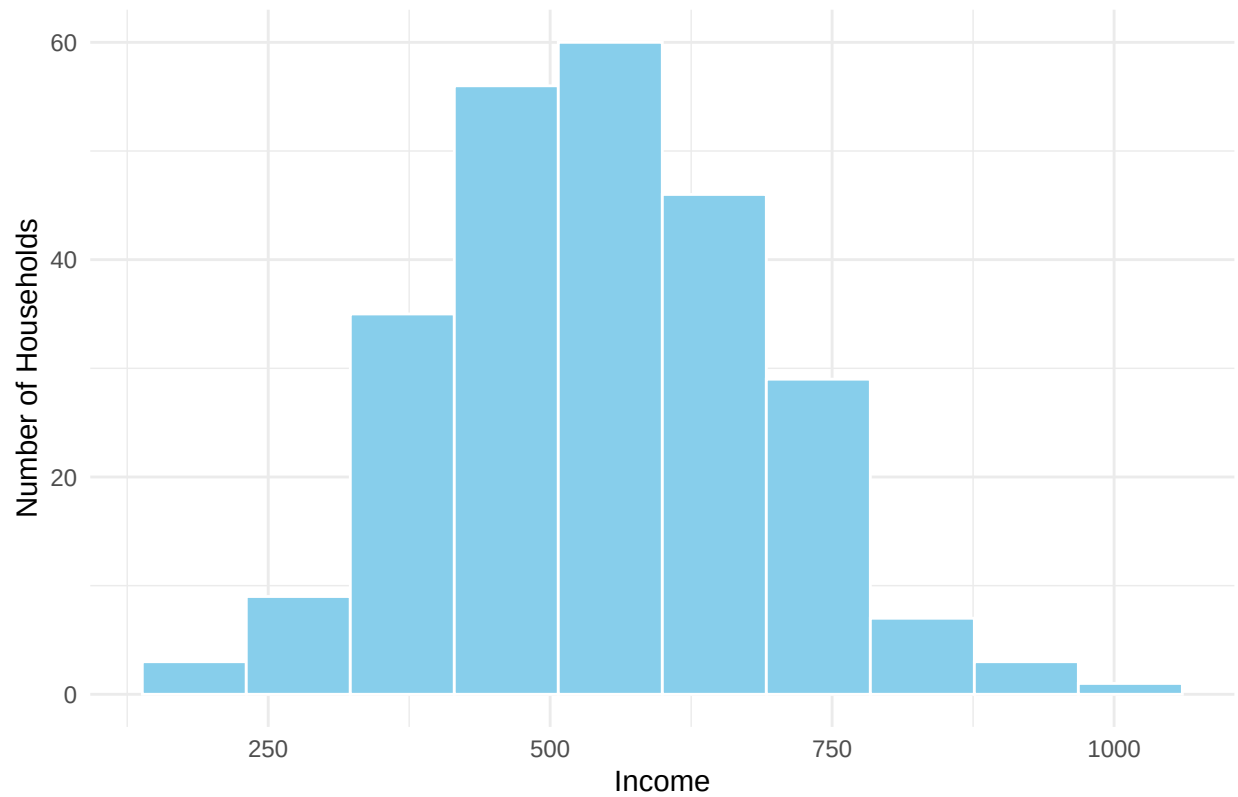
Pre-Treatment Income on Treated Households



```
# repeat for untreated households
untreated_inc_pre <- df_data$baseline_income[df_data$farms_trt == 0]

# plot income of untreated households
ggplot(data.frame(untreated_inc_pre), aes(x = untreated_inc_pre)) +
  geom_histogram(fill = "skyblue", color = "white", bins = 10) +
  labs(title = "Pre-Treatment Income on Untreated Households",
       x = "Income",
       y = "Number of Households") +
  theme_minimal()
```

Pre-Treatment Income on Untreated Households



```
# BALANCE TEST
# run regression on each covariate (week 2 TA session)

regressions <- df_data %>%
  select("baseline_income",
         "baseline_agexp",
         "baseline_hhsize",
         "baseline_livestock") %>%
  lapply(function(x) lm(x ~ farms_trt, data = df_data))

# pull the regression coefficient and p-value
balance_table <- regressions %>%
  sapply(function(x) coef(summary(x))[c(2,8)]) %>%
  t()

# format the table output
balance_table %>%
  kable(col.names = c("Differences in means", "p-value")) %>%
  kable_styling(position = "center", font_size = 8.5, latex_options = "hold_position")
```

	Differences in means	p-value
baseline_income	11.3382774	0.3911449
baseline_agexp	0.9806077	0.6031304
baseline_hhsize	0.0242564	0.7922575
baseline_livestock	-0.0646570	0.1433270

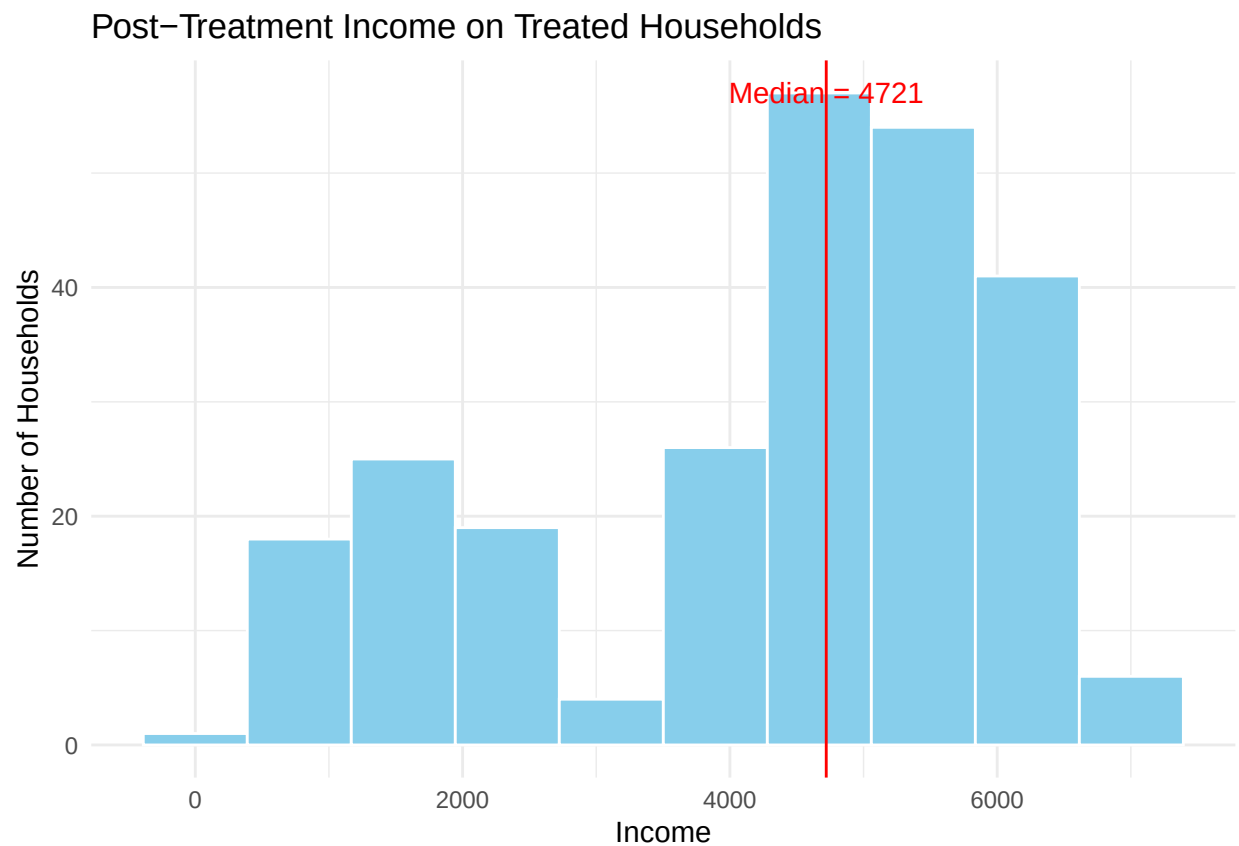
6.

```
# create histogram of post treatment incomes

# define an object that holds earnings for treated households
treated_inc <- df_data$endline_income[df_data$farms_trt == 1]

# compute the median treated income
med_treat_inc <- median(treated_inc, na.rm = TRUE)

# plot income of treated households
ggplot(data.frame(treated_inc), aes(x = treated_inc)) +
  geom_histogram(fill = "skyblue", color = "white", bins = 10) +
  geom_vline(xintercept = med_treat_inc, color = "red") +
  annotate("text",
    x = med_treat_inc,
    y = Inf,
    label = paste("Median =", round(med_treat_inc, 2)),
    vjust = 2,
    color = "red") +
  labs(title = "Post-Treatment Income on Treated Households",
    x = "Income",
    y = "Number of Households") +
  theme_minimal()
```



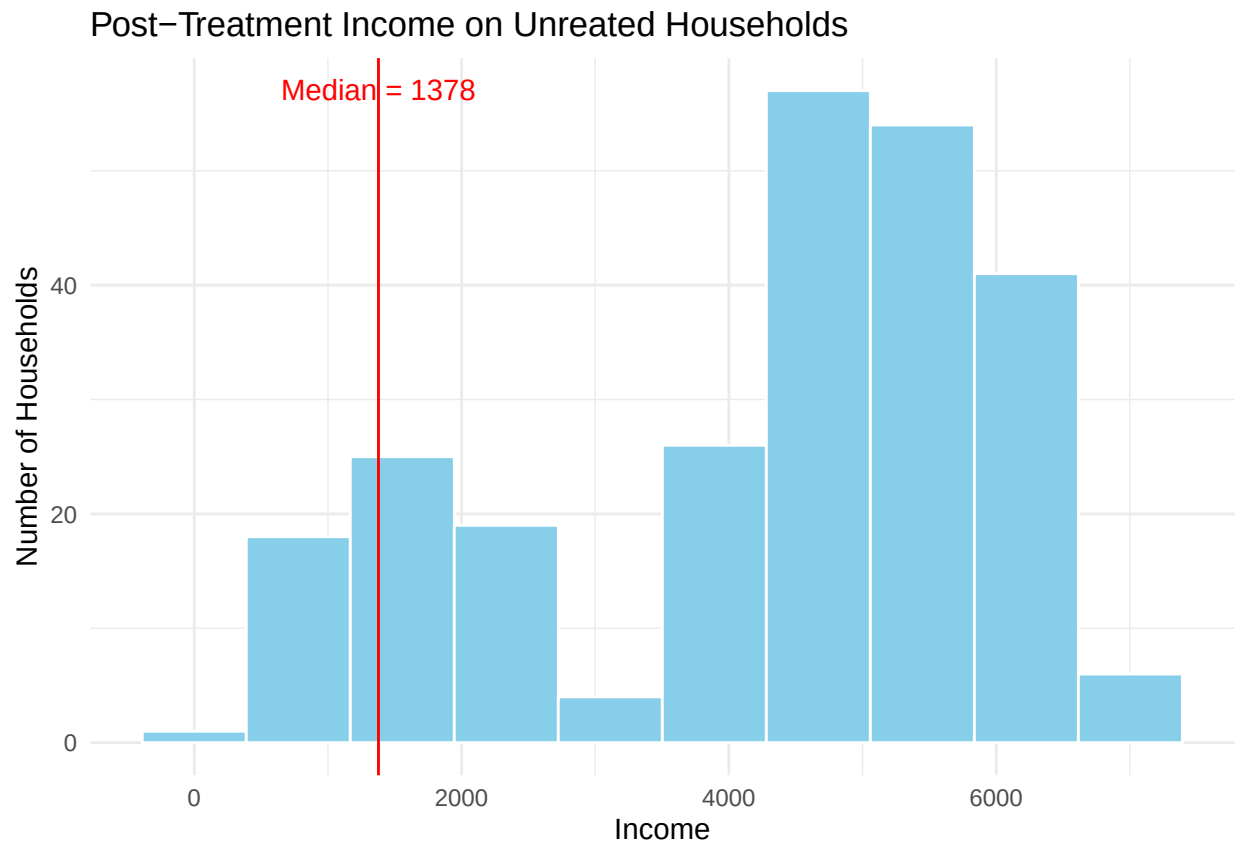
```
# define an object that holds earnings for untreated households
untreated_inc <- df_data$endline_income[df_data$farms_trt == 0]
```

```

# compute the median untreated income
med_untreat_inc <- median(untreated_inc, na.rm = TRUE)

# plot income of untreated households
ggplot(data.frame(treated_inc), aes(x = treated_inc)) +
  geom_histogram(fill = "skyblue", color = "white", bins = 10) +
  geom_vline(xintercept = med_untreat_inc, color = "red") +
  annotate("text",
    x = med_untreat_inc,
    y = Inf,
    label = paste("Median =", round(med_untreat_inc, 2)),
    vjust = 2,
    color = "red") +
  labs(title = "Post-Treatment Income on Untreated Households",
    x = "Income",
    y = "Number of Households") +
  theme_minimal()

```



The difference in average income (\$/month) between the treatment and control group is $4721 - 1378 = 3343$.

```

# estimate the ATE

base_rg <- lm(endline_income ~ farms_trt,
  data = df_data)

summary(base_rg)

```

```
Call: lm(formula = endline_income ~ farms_trt, data = df_data)
```

```
Residuals: Min 1Q Median 3Q Max -3916.4 -754.5 150.9 959.6 3087.6
```

```
Coefficients: Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) 1521.76 88.25 17.24 <2e-16 farms_trt 2736.63 124.56 21.97 <2e-16 — Signif. codes: 0 ‘  
0.001 ’ 0.01 ’ 0.05 ’ 0.1 ’ 1
```

```
Residual standard error: 1393 on 498 degrees of freedom Multiple R-squared: 0.4922, Adjusted R-squared:  
0.4912 F-statistic: 482.7 on 1 and 498 DF, p-value: < 2.2e-16
```

The interpretation of the coefficient is that access to the microcredit increased income by 2736.63 U.S. dollars per month. This result is statistically significant. This is 606.37 \$/month less of an increase than the simple difference in means. The effect is larger relative to the mean in the control group (1378).

7. This is not a good idea because they're essentially proposing that we control for a factor that (may) directly impact the outcome variable. If agricultural investment is the mechanism by which microcredit increases income, then controlling for agricultural investment will be akin to controlling for our outcome variable. This would mask the treatment effect and significantly bias our estimate.

The (bad) regression suggests that the treatment is not effective (coefficient of 7.1 and p-value of 0.52), but that post-treatment agricultural expenditure is attributed to a \$2.40 monthly increase in income (p-value of 0, highly significant results). While the regression from question 6 suggests a large, statistically significant treatment effect, this new regression suggests a small and statistically insignificant treatment effect.

After controlling for baseline agricultural investment, we get a 2,734 coefficient for the treatment variable (p = 0, statistically significant). We also see a small (1.87) and statistically insignificant (p = 0.528) effect from pre-treatment agricultural investment. This new coefficient is nearly identical to our estimate from question 6 (2734 vs. 2737), and both this regression and the Q6 regression p-values of approximately zero.

Word count: 188

```
# regress income on treatment and endline amount of agricultural investment
```

```
reg_end_agexp <- lm(endline_income ~ farms_trt + endline_agexp,  
  data = df_data)
```

```
summary(reg_end_agexp)
```

```
##
```

```
## Call:
```

```
## lm(formula = endline_income ~ farms_trt + endline_agexp, data = df_data)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -1329.97 -678.70  -58.96   638.03  2342.45
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  1.344e+03  5.109e+01  26.316  <2e-16 ***
```

```
## farms_trt     7.103e+01  1.104e+02   0.643    0.52
```

```
## endline_agexp  2.420e+00  7.626e-02  31.732  <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 801.3 on 497 degrees of freedom
```

```
## Multiple R-squared:  0.8322, Adjusted R-squared:  0.8315
```

```
## F-statistic: 1232 on 2 and 497 DF, p-value: < 2.2e-16
```

```
# regress income on treatment and baseline ag investment

reg_bl_agexp <- lm(endline_income ~ farms_trt + baseline_agexp,
                  data = df_data)

summary(reg_bl_agexp)

##
## Call:
## lm(formula = endline_income ~ farms_trt + baseline_agexp, data = df_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3881.9  -736.7   171.2   954.3  3101.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1384.352    234.814   5.896  6.9e-09 ***
## farms_trt     2734.797    124.665  21.937 < 2e-16 ***
## baseline_agexp    1.871     2.963   0.632   0.528
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1393 on 497 degrees of freedom
## Multiple R-squared:  0.4926, Adjusted R-squared:  0.4906
## F-statistic: 241.3 on 2 and 497 DF,  p-value: < 2.2e-16
```

8. In light of this new information, I estimated the intent-to-treat (ITT) estimate in question 6. The ITT estimate measures the effect of offering treatment, which includes non-compliance. The ITT differs from the ATE in that the ATE measures the actual effect of the treatment while the ITT measures the effect of offering treatment. The ITT is always less than the ATE.

9. In math: ATT (average treatment effect on the treated) = $E[Y_i(1) - Y_i(0) | D_i = 1]$ Using data on treatment assignment and actual treatment (where the treatment is the actual use of a microcredit loan), we can estimate the average treatment effect on the treated (ATT). In this case, the ATT is the estimated average effect of a microcredit on income specifically for the population of households who have access to microcredits. This allows us to estimate the effect of the actual microcredit without relying on an assumption of compliance.

Our ITT estimate is 3618 \$/month (see below). The regression suggests that, for households with access to microcredits, the use of a microcredit will result in an additional (compared to baseline) \$3,618 per month. This result exhibits high statistical significance with a p-value of about zero. This new estimate is nearly \$1,000 higher than the one from question 6. This suggests that microcredits are highly effective for the types of households that have access to them, and lack of compliance within the treatment group negatively biased our average treatment effect estimate.

```
# source: https://www.statology.org/subset-data-frame-in-r/

# make a new df with only those assigned treatment
df_treated <- subset(df_data, farms_trt == 1)

# run a regression on microcredit usage
itt_reg <- lm(endline_income ~ farms_got_loan,
              data = df_treated)

summary(itt_reg)
```

```
##
## Call:
## lm(formula = endline_income ~ farms_got_loan, data = df_treated)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1315.24  -635.40   -14.24    610.10   2121.76
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1605.9      96.4    16.66  <2e-16 ***
## farms_got_loan    3618.3     112.6    32.14  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 789.1 on 249 degrees of freedom
## Multiple R-squared:  0.8057, Adjusted R-squared:  0.805
## F-statistic: 1033 on 1 and 249 DF, p-value: < 2.2e-16
```

10. If they are correct, then the estimate in question 6 underestimates the treatment effect. If the control households have increased income as a result of their neighbors receiving the microcredit, then the true difference between the effects of receiving the microcredit and not receiving the microcredit is higher than the calculated difference from question 6. The income of control households is an overestimate, which results in downward bias for the estimated treatment effect. The estimated treatment effect in question 6 is the lower bound for the true treatment effect because spillovers make the average outcome higher than without treatment.